# PROGRAM 3 – Creating and Managing Branches

Write the commands to stash your changes,switch branches,and then apply the stashed changes.

## Aim

To temporarily save and retrieve changes using the Git Stash command.

## Procedure

**Step 1: Create a new file with some content.**

> echo "This is a new content for stashing" > stash.txt

**Explanation:** Creates a file named stash.txt and writes the given content into it.

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**Step 2: Display the files in the current directory.**

> ls

**Explanation:** Displays the files and folders present in the current directory.

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**Step 3: Add the changes to the staging area.**

> git add .

**Explanation:** Adds all the new and modified files to the staging area.

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**Step 4: Switch to the feature branch.**

> git checkout feature_branch

**Explanation:** Switches the current working branch from main to feature_branch.

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**Step 5: Switch back to the main branch.**

> git checkout main

**Explanation:** Switches the current branch back to main.

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**Step 6: Temporarily save the changes using Stash.**

> git stash save

**Explanation:** Temporarily stores the changes and gives you a clean working directory.

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**Step 7: View the saved stashes.**

> git stash list

**Explanation:** Displays all the changes that have been temporarily saved in the stash.

Example:

stash@{0}: WIP on main: ...

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**Step 8: Apply a saved stash.**

> git stash apply stash@{0}

**Explanation:** Restores the changes from the selected stash without removing the stash from the stash list.

**Note:** Use the stash name shown by git stash list. For example, stash@{0}.

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**Step 9: Commit the restored changes.**

> git commit -m "stash applied"

**Explanation:** Saves the restored changes as a new commit.

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**Step 10: Remove the selected stash.**

> git stash drop stash@{0}

**Explanation:** Deletes the selected stash from the stash list.

**Note:** Replace stash@{0} with the stash name you want to remove.

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**Step 11: Apply and remove the latest stash.**

> git stash pop

**Explanation:** Applies the latest saved stash and removes it from the stash list.

### Important Difference

git stash apply → Apply stash, but keep it in the stash list.

git stash pop → Apply stash and remove it from the stash list.

git stash drop → Remove a stash without applying it.

# PROGRAM 4 – Collaboration and Remote Repositories

Clone a remote Git repository to your local machine.

**Aim**

To copy an existing GitHub repository to the local computer using the "git clone" command.

**Procedure**

**Step 1:** Open the required DBMS repository on GitHub.

Go to the GitHub repository containing the DBMS code.

Click:

Code → HTTPS → Copy the repository link

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**Step 2:** Clone the GitHub repository.

> git clone <GITHUB-REPOSITORY-LINK>

Explanation: Downloads a copy of the GitHub repository, including its files and Git history, to the local computer.

Example

git clone https://github.com/username/DBMS.git

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**Step 3:** Enter the downloaded repository.

> cd DBMS

Explanation: Moves into the newly downloaded "DBMS" repository folder.

«Note: Replace "DBMS" with the actual name of the repository created on your computer.»

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Complete Commands – Copy & Paste

git clone <GITHUB-REPOSITORY-LINK>
cd <REPOSITORY-NAME>

Example:

git clone https://github.com/username/DBMS.git
cd DBMS

After this, you can use Git commands inside the cloned repository.